

Computation in Engineering Physics

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Problem:

We want to calculate fuel expenditures of a spacecraft when changing orbits and transferring to different celestial bodies, such as the **Moon** and/or Mars. We also want to keep track of any time dilation that the spacecraft may experience.

Formulating the Problem Mathematically Using:

- Differential equations
- Optimizing Functions
- Linear Algebra-related functions

Work division (in order from most contribution to least):

Frank - Orbital Calculations, Time Dilation, Python Programming

Johann - Python Programming, Time Dilation, Orbital Calculations

Will - Python Programming, Orbital Calculations, Time Dilation

Planned Methods:

We have to consider the mass of the spacecraft, the mass of the fuel, the specific impulse of our spacecraft's engine, as well as our starting orbit around the Earth and the gravitational pull of the other celestial bodies.

These situations will require the utilization of differential equations developed from the *Two-Body Problem* (Changing a spacecraft's orbit solely around Earth) and restricted cases of the *Three-Body Problem* (Sending a spacecraft on a transit trajectory to the Moon)

We will visualize the expenditure of fuel, our delta v, altitude data, etc., via *Matplotlib*. In addition, we will use *WebVPython* to see a visual representation of our code in order to see if we're on the right track.

Software and Libraries Used:

- Development Environments:
 - Spyder (for main program)
 - WebVPython (to visualize orbital trajectories)
- Libraries:
 - Scipy
 - Numpy
 - Matplotlib